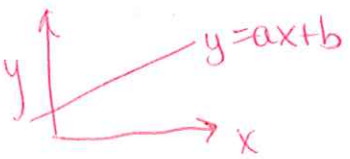
September 29 30

With regression models, we can explore relationships between variables, or predict the value of one variable based on the value of another variable.

regression models let us
* describe relationships
* make predictions

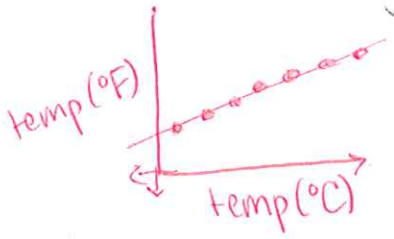
And we're focusing on linear regression, so specifically will study linear relationships between variables.

We have some variable X, some variable Y, and represent the relationship between them with a line.



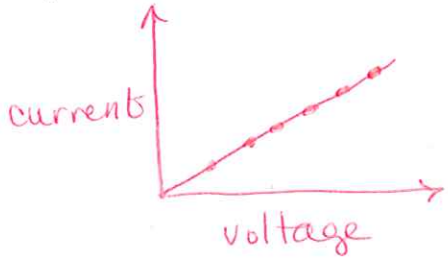
eqn of the line is $ax+b$, where "a" is the slope, "b" is the intercept.

If two quantities have a perfectly linear relationship, we can determine the eqn of that line by observing data (any observations of those variables will fall perfectly on a line)



eg temperature in Celsius and Fahrenheit

or



voltage & current across a resistor

But in most real-world settings, the variables we're interested in don't perfectly follow a linear relationship.

If we plot some observed data, the points aren't in a perfect line.



This connects to some actual data we'll look at later, about economic voting behavior. When economic conditions are good, people tend to have more favorable views toward the incumbent political party, and that party tends to get a bigger % share of the votes in an election.

We have a predictor, X, which is the economic growth, and a response Y, the incumbent vote share.

X = "predictor" (growth)

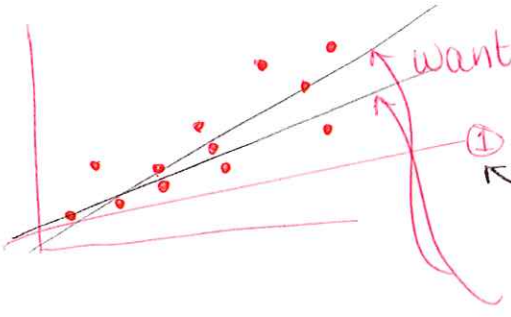
Y = "response" (vote share)

The relationship looks sort of linear. If we wanted to draw a line through those points to approximately describe the relationship, how would we do that?

≈ linear relationship

what is the line?

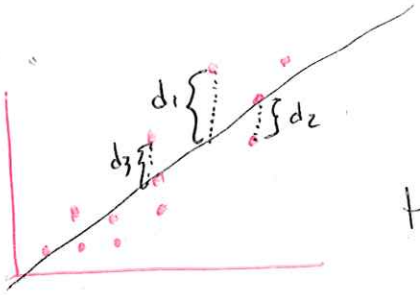
We want a line that's "close" to the data points on the scatter plot.



← qualitatively, we know this line isn't "close".

but what about these two lines? How do we decide which one of these is better, ie closer to the points?

how to define "close"? The most common way to quantify this, for linear regression, is to say: we want to minimize the vertical distances from the points to the line.



we want that vertical distance to be small across the whole set of points.

idea: sum of d_i's ? Conceptually, we want the sum of those vertical distances to be small.

But we don't want pos. & neg. distances to cancel each other out, so we square them, and want to draw a line that

don't want pos & neg to cancel
↳ square them, want small $\sum_{i=1}^{n} (d_i)^2$

minimizes the sum of the squared distances.

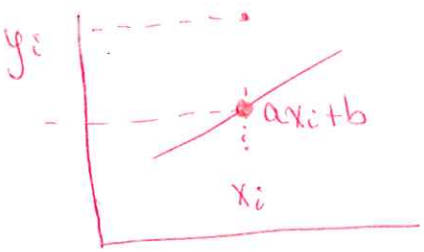
Let's introduce some notation: our data consists of pairs of observations of x and y:

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

If the equation of a line is $f(x) = ax + b$, then the point on the line for $x = x_i$ is

line: $f(x) = ax + b$ point on line: $(x_i, ax_i + b)$

Then at $x = x_i$, the vertical distance between the data point (x_i, y_i) and the corresponding point on the line is



$$(y_i - (ax_i + b))$$

and the sum of squared distances that we want to minimize is

want small $\sum_{i=1}^{n} (y_i - (ax_i + b))^2$

For our purposes, that is how we define the "best" line through the set of points. We want the line that minimizes the sum of squared distances. Also commonly referred to as the "line of best fit" or "least squares regression line"

line that minimizes $(-)^2$: "line of best fit"

"least squares regression line"

For a given data set, how do we find that line? Let's look at an example.

~ continued in notebook ~

As we've been discussing, the relationship between x and y won't be perfectly linear for most problems we care about.

Instead, we assume the true relationship between X and Y looks like

$$Y = \beta_0 + \beta_1 X + \text{error}$$

Instead of data points falling exactly on a line, they're spread out around the line - each point is away from the line by some error term.

To relate this to statistics, we consider the error terms to be random

error terms are random

~~On each observation comes from a model~~

Pairs of x and Y values have the relationship

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \longrightarrow \text{random error}$$

In this model, the random error terms have expected value zero. (assumption)

$E(\varepsilon_i) = 0$ for all i

Let's use an example to discuss this. Suppose we're interested in the relationship between a student's SAT score and their GPA in their 1st yr of college.

X = SAT score Y = freshman GPA

Assume for now that the true relationship is

GPA $= 1 + 0.002 * SAT + \varepsilon$

suppose

For any particular student, their 1st yr GPA equals 1 + 0.002* their SAT score, plus some random error.

What is the GPA of a student who scored 1200 on the SAT?

GPA of student w/ 1200 SAT score?

$X_i = 1200$

$Y_i = 1 + 0.002 * 1200 + \varepsilon_i$

$Y_i = 3.4 + \varepsilon_i$

There is also a "population distribution" of GPAs for students who scored 1200 on the SAT. Because of the random error term, "GPA of student w/1200 SAT score" is a random variable.

ε random → Y random

We could go out and collect the GPAs of multiple students with that SAT score, and they would be different, but would be close to 3.4 on average.

$$E(Y|X=1200) = E(1 + 0.002*1200 + \varepsilon)$$
$$= E(3.4 + \varepsilon)$$
$$= 3.4 + E(\varepsilon)$$
$$= 3.4$$

The expected GPA of a student who scored 1200 on the SAT is 3.4, because the expected value of the error term is zero.

More generally, the expected y value for a particular value of X is given by the corresponding point on the line.

generally: $E(Y|X=x) = 1 + 0.002X$

"typical" value of Y when $X=x$ is $\beta_0 + \beta_1 X$

There is a conceptual point to address here. In this example, we assumed there was a true relationship between GPA and SAT score, and that it was GPA = 1 + 0.002 SAT + error

this example: assumed true population relationship

$$GPA = 1 + 0.002\,SAT + error$$

In practice, we don't get to know this true relationship.

in practice: don't know true relationship

This is the same as our other problems involving an unknown population quantity, like not knowing the population % of drivers who wear seat belts.

didn't know p = % seatbelt users → estimated from data

In that example, there was some true population percentage, but we didn't know it, so we estimated it from data.

And in the GPA/SAT score example, in practice we won't know the true population relationship, but will estimate from data

regression

don't know true population X, Y relationship → estimate slope & intercept from data.

We can go collect a random sample of pairs of GPA & SAT scores, make a scatter plot, and fit a line to it.

observe $(x_1, y_1), \dots, (x_n, y_n)$ estimate $\hat{\beta}_0$, $\hat{\beta}_1$ ($\neq$ true β_0, β_1)

But the line that we fit to randomly sampled data won't be exactly equal to the true population values, just like sample proportions and sample means aren't exactly equal to their respective population values.

The regression line that we estimate from the data is:

estimated regression line: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

So for a particular data point (x_i, y_i), the corresponding point on the estimated regression line is

data (x_i, y_i) → Corresp point on line: $(x_i, \hat{\beta}_0 + \hat{\beta}_1 x_i)$

The vertical distance between that data point and the estimated regression line is called the "residual" for that data point.

residual $\hat{e}_i = y_i - \hat{y}_i$

$= y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$

This is not the same as the error term in the true, unknown population model.

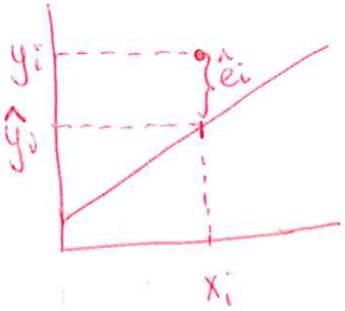
true population model $Y = \beta_0 + \beta_1 x + \varepsilon$

$\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i)$

unknown

We don't know how far away a data point is from the true population line (ε_i), only how far away it is from (the estimated regression line ($\hat{e}_i$), the residual).

The residuals are the same vertical distances between the points and the line that we discussed @beginning.



The least squares regression line minimizes the sum of the squared residuals.

least squares regression minimizes $\sum_{i=1}^{n} \hat{e}_i^2$

We aren't going to focus on how you show that mathematically, but we will state how to calculate the slope & intercept.

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$\hat{\beta}_1$ estimates β_1 ; $\hat{\beta}_0$ estimates β_0

When we talked about sample proportions, we used the binomial dist to confirm that the expected value of the sample proportion equals the true population proportion.

recall: $\mathbb{E}(\hat{p}) = p$ $\qquad \mathbb{E}(\bar{X}) = \mathbb{E}(X)$, μ

And likewise for sample means, the expected value of the sample mean was equal to the population mean.

This property, where the expected value of the estimator equals the true quantity we want to estimate, is called being "unbiased".

"unbiased": $\mathbb{E}(\hat{\theta}) = \theta$.

($\hat{\theta}$: estimator; θ: true value)

And the slope and intercept estimates are also unbiased

$\mathbb{E}(\hat{\beta}_0) = \beta_0$, $\mathbb{E}(\hat{\beta}_1) = \beta_1$

For any individual data set we go collect, our estimates of the slope and intercept won't be exactly correct, but we'd be correct on average. This method doesn't lead us to systematically guess too high ($\mathbb{E}(\hat{\beta}_1) > \beta_1$). doesn't lead us to always guess too low either ($\mathbb{E}(\hat{\beta}_1) < \beta_1$)

$$\mathbb{E}\left(\frac{\sum_i (x_i-\bar{x})(y_i-\bar{y})}{\sum_i (x_i-\bar{x})^2}\right) = \frac{\sum_i \mathbb{E}\left((x_i-\bar{x})(y_i-\bar{y})\right)}{\sum (x_i-\bar{x})^2}$$

$$= \frac{\sum_i (x_i-\bar{x})(\mathbb{E}(y_i)-\mathbb{E}(\bar{y}))}{\sum_i (x_i-\bar{x})^2}$$

$$\mathbb{E}(\bar{y}) = \mathbb{E}(\sum y_i)$$
$$= \sum \mathbb{E}(\beta_0+\beta_1 x_i+\varepsilon_i)$$
$$= \sum \beta_0+\beta_1 x_i+\mathbb{E}[\varepsilon_i]^0$$
$$= \beta_0+\beta_1\bar{x}$$

$$= \frac{\sum (x_i-\bar{x})(b_0+b_1 x_i-(b_0+b_1\bar{x}))}{\sum (x_i-\bar{x})^2}$$

$$= \sum (x_i-\bar{x})(\beta_1 x_i-\beta_1\bar{x}) \Big/ \sum (x_i-\bar{x})^2$$

$$= \frac{\beta_1 \sum (x_i-\bar{x})^2}{\sum (x_i-\bar{x})^2}$$

$$= \beta_1$$

Like we did for sample proportions & sample means, we will eventually want to ask how CLOSE our estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are to the truth.

↳ we know the expected value

↳ what is the variance? (How far away do they tend to be?)

We will cover this in more detail later on, but the general idea that observing more data makes the estimates more precise will still be true (as was the case for sample proportions & means).